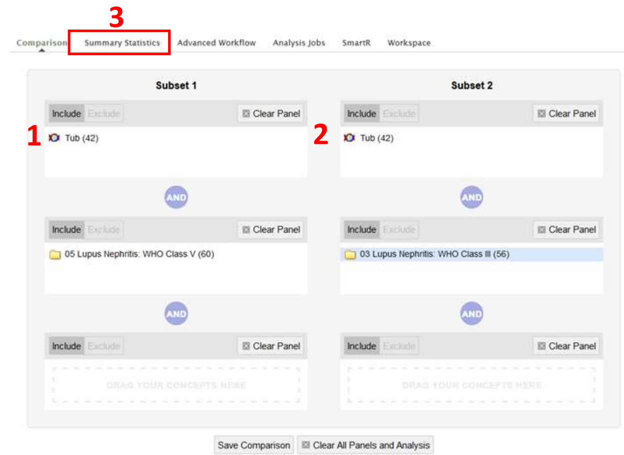
# CPROBE tranSMART Quick Start Guide
# Marker Selection (Differential Expression)

*A Marker Selection analysis generates a heatmap that visualizes the differences in expression from one sample set to another.*

## Step 1: Select sample sets

To run a Marker Selection analysis, set up two sample sets in the Subset 1 and Subset 2 boxes on the **Comparison** tab. Subset 1 (**1**) is the reference group, and Subset 2 (**2**) is the comparison group. The example shows Subset 1 with LN Class V and Tub samples, and Subset 2 with LN Class III and Tub samples.

Click on the **Summary Statistics** (**3**) tab to review the number of samples that will be included in the analysis. Confirm that the number of samples available for each subset is sufficient for running a meaningful analysis.

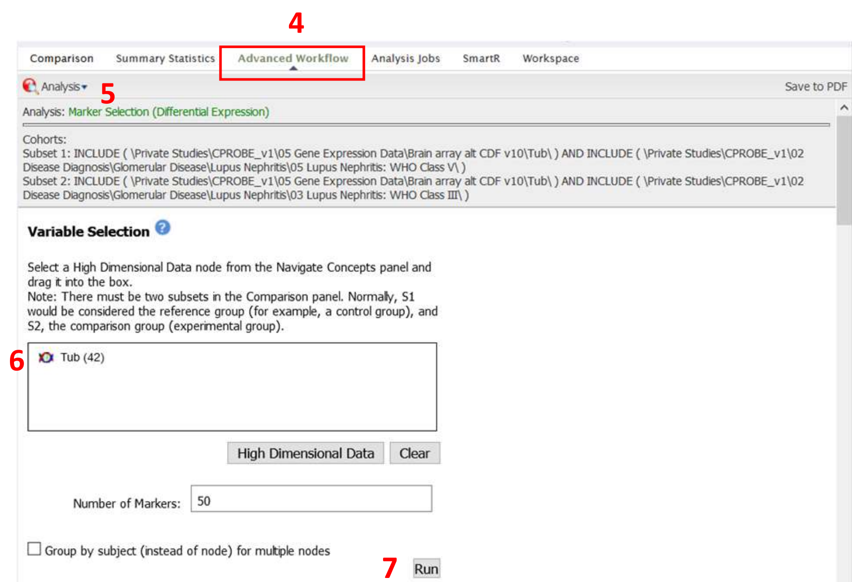



## Step 2: Set up analysis

Click on the **Advanced Workflow** tab (**4**) and select "Marker Selection (Differential Expression)" from the Analysis dropdown list (**5**).

Drag the high dimensional data node from the Navigate Concepts tree into the box (**6**). (Do NOT add a specific gene of interest, because the analysis will not run. If you would like to view a single probe, use the Heatmap analysis instead.)

Click the "Run" button (**7**). A popup box will show a progress bar as the data is compiled and analyzed by tranSMART. Some analyses, especially those with high dimensional data, may take several minutes to run.




## Step 3: Review results

A heatmap and a table of top markers will appear underneath the variable selection box. The subsets are delineated by the orange and yellow bars at the top of the heatmap (**8**).